

Control Narrative

WTS-100 Water Treatment and Transfer Skid

Describes how the skid behaves. The PLC program is written from this document.

1. What the skid does

Water is pumped from a raw water tank into a treatment tank. It is heated, a chemical is added, it is held for a set time, then it is pumped out. The cycle then repeats.

1.1 Equipment

Tag	Description
T-101	Raw water tank, 3000 mm
P-101A / P-101B	Raw water pumps, 3 kW. One runs, one is the spare.
XV-101	Inlet isolation valve. Spring closed.
FCV-101	Inlet flow control valve. Air to open.
T-102	Treatment tank, 2500 mm
HTR-101	Immersion heater, 6 kW
T-103	Chemical tank, 200 L
P-103	Dosing pump
XV-102	Discharge isolation valve. Spring closed.
P-102	Discharge pump

1.2 Signal count

7 analog inputs, 4 analog outputs, 14 digital inputs, 8 digital outputs. Devices are listed in WTS-100-II-001, the instrument index.

2. Modes

The operator selects the mode with switch HS-01.

Off. Everything is off. Both valves close.

Hand. The operator can run each device one at a time from the HMI. Used for maintenance and for checking loops. All interlocks in section 5 still apply.

Auto. The sequence in section 4 runs. The operator can Start, Hold and Abort.

Auto can only be selected from IDLE. Selecting Hand while running stops the skid first, in the order given in section 4.8.

3. Control loops

3.1 Level, LIC-102

Holds the level in T-102. It does not move the valve itself. It tells the flow loop what flow to run.

Reads	LT-102, 0 to 2500 mm
Setpoint	1800 mm. Operator range 500 to 2200 mm.
Output	Flow setpoint to FIC-101
Tuning	Kp 1.2, Ti 240 s
Deadband	10 mm

No derivative term is used.

3.2 Flow, FIC-101

Holds the inlet flow by moving valve FCV-101.

Reads	FT-101, 0 to 50 m3/h
Setpoint	From LIC-102 in Auto. Set locally in Hand.
Output	FCV-101, 0 to 100 percent
Tuning	Kp 0.8, Ti 12 s
Cutoff	Reading forced to zero below 1.0 m3/h

If FIC-101 is switched to local, LIC-102 follows the local setpoint. This stops the valve jumping when cascade is switched back on.

3.3 Temperature, TIC-102

Holds the temperature in T-102 by adjusting heater power.

Reads	TT-102, 0 to 100 degC
Setpoint	45 degC. Operator range 20 to 70 degC.
Output	HTR-101, 0 to 100 percent
Tuning	Kp 4.0, Ti 180 s, Td 20 s
Deadband	0.5 degC

The output is limited to 0 to 100 percent. While it sits at a limit, the integral term is held.

There is no cooling. The loop can only add heat.

3.4 Dosing, FFIC-103

Adds chemical in proportion to the water flow.

$$\text{dose setpoint (L/h)} = \text{water flow (m3/h)} \times \text{ratio (L/m3)}$$

Reads	FT-103, 0 to 500 L/h
Ratio	8.0 L/m3. Operator range 2.0 to 20.0.
Output	P-103 speed, 0 to 100 percent
Tuning	Kp 0.6, Ti 8 s
Inhibit	Dosing stops when water flow is below 2.0 m3/h

4. Sequence

One state is active at a time. Any interlock in section 5 sends the sequence to TRIP from any state.

4.0 IDLE

Everything off. Both valves closed.

Go to FILL when: Start is pressed, no trip is active, and LT-101 is above 400 mm.

4.1 FILL

Open XV-101. Start the duty pump. Hold flow at 30 m³/h. Heater off. Dosing off.

Go to HEAT when LT-102 reaches 1200 mm.

Timeout 900 s. Raise alarm A-009 and go to DRAIN.

4.2 HEAT

Put LIC-102 in cascade at 1800 mm. Start the temperature loop. Start dosing.

Go to HOLD when TT-102 stays within 0.5 degC of setpoint for 60 s.

Timeout 1800 s. Raise alarm A-009 and go to DRAIN.

4.3 HOLD

Keep level, temperature and dosing running. Run the residence timer.

Go to DISCHARGE when the timer reaches 600 s. Operator range 300 to 3600 s.

4.4 DISCHARGE

Heater off. Dosing off. Raw pumps off. Close XV-101. Open XV-102. Run P-102 at 60 percent.

Go to DRAIN when LT-102 falls to 300 mm.

Timeout 600 s. Raise alarm A-009.

4.5 DRAIN

Run P-102 at 100 percent.

Go to IDLE when LT-102 stays at or below 50 mm for 10 s. Close XV-102.

4.6 PAUSE

Entered when the operator presses Hold. Timers stop. FCV-101 closes. Heater goes to 0 percent. Pumps stop.

Press Start to go back to the state it came from.

4.7 TRIP

Entered on any interlock in section 5.

All pumps off. Heater off. FCV-101 closed. XV-101 and XV-102 closed.

TRIP is latched. It clears only when the cause has gone and the operator presses Reset.

4.8 Controlled stop

When the operator presses Abort, devices stop in this order:

1. Heater
2. Dosing
3. Pumps
4. Valves close

A trip is different. A trip drops everything at once.

5. Interlocks

Safety contacts are wired normally closed. Closed means healthy. A broken wire opens the circuit and trips.

Interlocks are checked at the start of every scan, before the sequence and the loops.

#	Trips on	Sensed by	Result	Reset
I1	Emergency stop pressed	ESD-01	Everything off. A safety relay also drops the contactors without the PLC.	Manual
I2	Raw tank empty	LSLL-101	Raw pumps off	Auto, after 10 s
I3	Treatment tank too full	LSHH-102	Raw pumps off, XV-101 and FCV-101 closed	Auto
I4	Too hot, 85 degC	TSHH-102	Heater off. Also wired in series with the heater contactor.	Manual
I5	Level below 800 mm	LT-102	Heater blocked	Auto
I6	Both raw pumps faulted	XA-101A and XA-101B	TRIP	Manual
I7	Analog signal bad for 30 s	FB_AIN_SCALE	TRIP if LT-102 or TT-102	Manual
I8	Level not falling while the discharge pump runs, 20 s	LT-102	P-102 off, XV-102 closed	Manual

5.1 How interlocks are applied

POU_SAFETY does not switch any device on. It only gives permission. The rest of the program has to ask for that permission first.

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PERM_HEATER := ESD_OK AND TSHH_OK AND LT_VALID AND TT_VALID
              AND (LEVEL >= 800) AND NOT TRIPPED

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The heater permission uses the measured level, not the sequence state. So a fault in the sequence cannot switch the heater on in an empty tank.

I1 and I4 are also wired in hardware. They work whether or not the PLC is running.

6. Alarms

Tag	Alarm	Priority	Reset
A-001	Emergency stop	Critical	Manual
A-002	Over temperature	Critical	Manual
A-003	Both raw pumps failed	Critical	Manual
A-004	Treatment tank too full	High	Auto
A-005	Raw tank empty	High	Auto
A-006	Duty pump failed, standby started	High	Manual
A-007	Analog signal bad	High	Auto
A-008	Signal not changing	Medium	Manual
A-009	Sequence timeout	Medium	Manual
A-010	Temperature more than 5 degC from setpoint	Medium	Auto
A-011	Pump feedback does not match command	Medium	Manual
A-012	Dosing more than 20 percent off setpoint	Low	Auto

The alarm system records which alarm came first. One fault often sets off several alarms, and the first one is usually the real cause.

Horn ALM-H sounds on a new alarm and is silenced by HS-04. Beacon ALM-B flashes until the alarm is acknowledged, then stays on while the alarm is still active.

7. Reference documents

Number	Title
WTS-100-CN-001	Control narrative (this document)
WTS-100-DB-001	Design basis
WTS-100-II-001	Instrument index
WTS-100-PID-001	P&ID
WTS-100-CAL-001	Signal conditioning and calibration
WTS-100-FAT-001	Factory acceptance test